

Supplementary material for ‘Noisy calibration, latent coverage and the location of the mean’

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Equation, theorem and section numbers without the prefix S refer to the main paper. Numbers that are not proved are labelled as numerical, and the precision of each computation is stated.

S1 Proofs omitted from the main paper

Proof of Proposition 1. Since $\mathcal{F} \subset \mathcal{L}$ it suffices to bound the supremum over $\mathcal{L}$. Let $W \in \mathcal{L}$ satisfy $E\{g_x(W)\} \geq p$ and let $s < Q_q(|W|)$, so that $\text{pr}(|W| < s) \leq q - \gamma$ for some $\gamma > 0$. Choose $M \geq s$ with $g_x(w) \leq p$ for $|w| \geq M$, and let W_M be W conditioned on $|W| \leq M$, which is log-concave. With $\varepsilon_M = \text{pr}(|W| > M)$, the conditioning removes only mass on which the kernel is at most p , so

$$E\{g_x(W_M)\} \geq \frac{p - p\varepsilon_M}{1 - \varepsilon_M} = p, \quad \text{pr}(|W_M| < s) \leq \frac{q - \gamma}{1 - \varepsilon_M} < q$$

for M large. On $K = [-M, M]$ the constraint function $g_x - p$ is continuous, and $\mu \mapsto -\mu(-s, s)$ is linear and upper semicontinuous for weak convergence because $(-s, s)$ is open. By the localization theorem (Kannan et al., 1995; Fradelizi & Guédon, 2004), the supremum of this functional over the log-concave probability measures on K that satisfy $\int (g_x - p) d\mu \geq 0$ is attained at a point mass or at a law with log-affine density on a segment. Hence some member of $\mathcal{F}$ is feasible and has $\text{pr}(|W| < s) < q$, that is, $Q_q(|W|) \geq s$. Since $s < Q_q(|W|)$ was arbitrary the claim follows. Only continuity of the kernel and its decay at infinity were used, so the argument applies to the average kernel $\bar{g}$ of § 4.2. $\square$

For $\beta > 0$ and $\sigma = x^{1/2}$, integration by parts gives

$$\int_a^b e^{\beta w} \Phi\left(\frac{c - w}{\sigma}\right) dw = \frac{1}{\beta} \left[e^{\beta w} \Phi\left(\frac{c - w}{\sigma}\right) \right]_a^b + \frac{e^{\beta c + \beta^2 \sigma^2 / 2}}{\beta} \left\{ \Phi\left(\frac{b - c - \beta \sigma^2}{\sigma}\right) - \Phi\left(\frac{a - c - \beta \sigma^2}{\sigma}\right) \right\}, \quad (\text{S1})$$

so the constraint is a combination of normal distribution functions. For fixed slope and length, the translations c of a member of $\mathcal{F}$ that satisfy the constraint form an interval, because $c \mapsto \text{pr}(|W + c + e| \leq 1)$ is log-concave (Prékopa, 1973), and $c \mapsto \text{pr}(|W + c| \leq s)$ is log-concave, so $c \mapsto Q_q(|W + c|)$ is quasi-convex and is maximized at an end of that interval. Each end is found by a root-finding in c , and the quantile by a second root-finding, using (S1). This two-dimensional computation over slope and length, and an independent differential-evolution search over the three-parameter family, agree to 1.5×10^{-13} at all 73 tabulated values of $x \geq 0.006$ (numerical).

Proof of Lemma 1. Exponential tails are log-concave, which gives one inequality. Conversely, let $Y \in \mathcal{L}$ with $\text{pr}(Y + Z \leq 0) \geq p$. If $Y = y_0$ is a point mass then $y_0 \leq -\Phi^{-1}(p) = \lim_{u \rightarrow \infty} \{b_u(p) - u^{-1} \log(1/q)\}$. Otherwise $F = F_Y$ is continuous and $\log F$ is concave (Prékopa, 1973). Let $v = F^{-1}(q)$ and $G(y) = F(v + y)$. The point 0 is interior to $\{G > 0\}$, so the right derivative $u \geq 0$ of $\log G$ at 0 is a supergradient, and $G(y) \leq qe^{uy}$ for all y . If $u = 0$ then $G \leq q < 1$, which is impossible, so $u > 0$ and $G(y) \leq \min(qe^{uy}, 1)$,

which is the distribution function of $b_0 - E_u$ with $b_0 = u^{-1} \log(1/q)$. Hence $Y' = v + b_0 - E_u$ lies stochastically below Y and has q -quantile v , and

$$p \leq \text{pr}(Y + Z \leq 0) \leq \text{pr}(Y' + Z \leq 0) = m_u(v + b_0).$$

Since m_u is decreasing, $v \leq b_u(p) - b_0$. For monotonicity, let $q' > q$, let Y be feasible at (q', q') with $v = F_Y^{-1}(q')$, and put $\rho = (q' - q)/(1 - q)$. Then Y conditioned on $Y \geq F_Y^{-1}(\rho)$ is log-concave, satisfies the constraint at level $(q' - \rho)/(1 - \rho) = q$, because the removed mass has noisy constraint value at most one, and has q -quantile v . Point masses have value $-\Phi^{-1}(q') < -\Phi^{-1}(q) \leq c_q$. Finally, $c_q > 0$ because $v_q(u) = u^{-1} \log\{e^{u^2/2} \Phi(b_u - u) + e^{ub_u} \Phi(-b_u)\} = u/2 + o(u)$ as $u \rightarrow 0$, where $b_u = b_u(q)$ and $ub_u \rightarrow \log(1/q)$. $\square$

Proof of Theorem 1, lower bound. Fix $\epsilon > 0$ and choose u with $v_q(u) > c_q - \epsilon$. For $p' \in [q, (1 + q)/2]$, $b_u(p') \geq b_{\min} > -\infty$. Let $Y = b_u(p') - E_u$ and $W = 1 + x^{1/2}Y$. With $\tau = 2x^{-1/2}$,

$$\text{pr}(|W + x^{1/2}Z| \leq 1) = p' - \text{pr}(Y + Z < -\tau) \geq p' - \eta_x, \quad \eta_x = e^{-u(\tau + b_{\min})/2} + \Phi\{-(\tau + b_{\min})/2\},$$

by splitting the event $Y + Z < -\tau$ according to whether $Z < -(\tau + b_{\min})/2$. Take $p' = q + \eta_x$, which lies in $[q, (1 + q)/2]$ for small x . Since $\text{pr}(|W| \leq s) \leq F_W(s)$, $Q_q(|W|) \geq 1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1} \log(1/q)\}$. As $\eta_x \rightarrow 0$ and b_u is continuous, $\liminf_{x \rightarrow 0} \{R_{q,q}(x) - 1\}/x^{1/2} \geq c_q - \epsilon$. $\square$

Proof of Lemma 2. For $q > 1/2$, v_q is continuous on $(0, \infty)$, tends to 0 as $u \rightarrow 0$ and to $-\Phi^{-1}(q) < 0$ as $u \rightarrow \infty$, and is positive somewhere, so its maximizers form a nonempty compact subset of $(0, \infty)$ and κ_q^* is well defined. Let $a = \log(1/q)$ and $A = 1 - a > 0$. Let Y be nondegenerate with q -quantile v . Since $\log F_Y$ is concave, a supergradient $u > 0$ at v gives $F_Y(y) \leq \min\{qe^{u(y-v)}, 1\}$, the distribution function of $Y' = v + a/u - E_u$; so Y' lies stochastically below Y and has the same q -quantile. Then $E(Y') \leq E(Y) \leq -\kappa$ and $\text{pr}(Y' + Z \leq 0) \geq q$, so the replacement keeps both constraints. For the tail $b - E_u$ the constraints read $b \leq b_u(q)$ and $b \leq u^{-1} - \kappa$, so the largest quantile is $\min\{v_q(u), A/u - \kappa\}$. If $k_q(u) < \kappa$ this minimum is $A/u - \kappa$; since k_q is continuous and $k_q(s) \rightarrow \infty$ as $s \rightarrow 0$, some $s < u$ has $k_q(s) = \kappa$, where the value is $A/s - \kappa > A/u - \kappa$. So only u with $k_q(u) \geq \kappa$ matter, and there the value is $v_q(u)$. A point mass y_0 satisfying the constraints has $y_0 = E(Y) \leq -\kappa \leq 0 < L_q(\kappa)$, where positivity holds because $v_q(u) \sim u/2$ and $k_q(u) \sim A/u$ as $u \rightarrow 0$. Since $m_u(b) \geq \Phi(-b)$, $b_u(q) \geq -\Phi^{-1}(q)$, so k_q is bounded on $[u_0, \infty)$ for every $u_0 > 0$ and $k_q(u) \geq \kappa$ forces $u \rightarrow 0$ as $\kappa \rightarrow \infty$; moreover $ub_u(q) \rightarrow \log(1/q)$, so $k_q(u) = \{A + o(1)\}/u$, and $v_q(u) = u/2 + o(u)$ as $u \rightarrow 0$ (proof of Lemma 1). These expansions give $\kappa L_q(\kappa) \rightarrow A/2$: for large κ only small u are feasible, $u \leq \{A + o(1)\}/\kappa$, and $v_q(u) \leq \{1 + o(1)\}u/2$ with equality at the boundary. Monotonicity is clear, and $L_q(\kappa) = c_q$ while a maximizer of v_q is feasible. $\square$

Proof of Proposition 2. Near $\pm r_0$ the density f of W is smooth with $f' = \beta f$, and the ends of the support are at positive distance. The jumps of the density at the ends of the support contribute $O\{v^{1/2}\phi(d_0/v^{1/2})\}$ near $\pm r_0$, where $d_0 > 0$ is the distance from $\pm r_0$ to the ends, which is $o(v)$. Hence $H_v(r) = \text{pr}(|W + v^{1/2}Z| \leq r) = H_0(r) + v\{f'(r) - f'(-r)\}/2 + o(v)$ uniformly near r_0 , and the noisy quantile is $t(v) = r_0 - v\beta\{f(r_0) - f(-r_0)\}/[2\{f(r_0) + f(-r_0)\}] + o(v) = r_0 - (\beta v/2) \tanh(\beta r_0) + o(v)$. Scaling by $t(v)$ and writing $x = v/t(v)^2$, a continuous and increasing function of v near 0, gives the expansion. As $\beta \ell \rightarrow 0$, W/ℓ tends to the uniform law on $[-1/2, 1/2]$, whose q -quantile of $|W|/\ell$ is $q/2 < 1/2$. $\square$

Proof of Theorem 3, the limit. For the lower bound in the limit, take u with $k_q(u) \geq \kappa$ and the law $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$ of the lower bound in Theorem 1. Its mean is $1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1}\} \leq 1 - \kappa x^{1/2}$, because b_u decreases in its argument, and it is positive for small x ; hence $\liminf_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \geq v_q(u)$ for every such u .

For the upper bound, suppose that $x_n \rightarrow 0$, $h_n = x_n^{1/2}$, and that W_n is feasible for $R_{q,q}^{(b_{x_n})}(x_n)$ with $Q_q(|W_n|) - 1 \geq a_0 h_n$, where $L_q(\kappa) < a_0 < 1$. By (2), $\kappa h_n \leq 1 - |E(W_n)| \leq K h_n$ with $K = (1 - a_0^2)/(2a_0)$. Replacing W_n by $-W_n$ if necessary, $E(W_n) \geq 0$; let $Y_n = (W_n - 1)/h_n$, so that $-K \leq E(Y_n) \leq -\kappa$. The law of $T_n = Y_n + Z$ is in $\mathcal{B}$, with a smooth positive density of maximum M_n and distribution function F_n , and $F_n(0) \geq \Pr(|W_n + h_n Z| \leq 1) \geq q$ while $F_n\{E(T_n)\} \leq 1 - e^{-1}$ by the first part. So $q - 1 + e^{-1} \leq F_n(0) - F_n\{E(T_n)\} \leq M_n K$. For a law in $\mathcal{B}$ with smooth density g , maximum M and median m , the hazard $g/(1 - F)$ is nondecreasing, so $g(y) \leq 2g(m)\{1 - F(y)\} \leq 2g(m)$ for $y \leq m$, and g/F is nonincreasing, which gives the same for $y \geq m$; so $g(m) \geq M/2$; the tangent bounds for $\log F$ and $\log(1 - F)$ at m give $\Pr(|T - m| > y) \leq e^{-2g(m)y}$, hence $\text{var}(T) \leq 1/\{2g(m)^2\} \leq 2/M^2$. Thus $\text{var}(Y_n) \leq 2K^2/(q - 1 + e^{-1})^2$, the Y_n are tight and uniformly integrable, and along a subsequence $Y_n \rightarrow Y$ in distribution with $Y \in \mathcal{B}$, possibly a point mass, since $\mathcal{B}$ is closed under weak convergence (Saumard, 2019), $E(Y) \leq -\kappa$ and $\Pr(Y + Z \leq 0) = E\{\Phi(-Y)\} \geq q$. On $\{W_n \geq 0\}$, whose probability tends to one, $(|W_n| - 1)/h_n = Y_n$, so $a_0 \leq \{Q_q(|W_n|) - 1\}/h_n \rightarrow Q_q(Y) \leq L_q(\kappa)$ by Lemma 2, where the quantile converges because a law in $\mathcal{B}$ has a unique q -quantile or is a point mass. This contradicts $a_0 > L_q(\kappa)$, so $\limsup_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \leq a_0$ for every $a_0 \in (L_q(\kappa), 1)$; if $L_q(\kappa) \geq 1$ there is nothing to prove, because (2) gives $\delta < x^{1/2}$. $\square$

Ranks for Proposition 4

At $q = 0.9$ and $\delta = 0.05$ no rank satisfies $p_k \geq 0.901$ and $k \geq Kp_k + 1$ for $K \leq 28$, since even $k = K$ gives $p_K = 0.05^{1/K} < 0.901$. For every K from 29, where the rank is $k = K$, to 3000 a rank exists, and it is at most three ranks above the usual rank with $p_k \geq 0.9$; for instance 46 against 45 at $K = 46$ and 2731 against 2728 at $K = 3000$. The gap grows with K , to 10 at $K = 10^4$ (9060 against 9050), because the slack 10^{-3} becomes large relative to the spacing of the levels p_k . The function `certified_rank` returns this rank.

Proof of Proposition 3. $\bar{F}(T) < p_k$ exactly when $N = \#\{i : |V_i| < \bar{F}^{-1}(p_k)\} \geq k$. N is a sum of independent indicators whose success probabilities average p_k , and $k - 1 \geq Kp_k$. By Hoeffding (1956), $\Pr(N \leq k - 1)$ is smallest when all success probabilities are equal. Hence $\Pr(N \geq k) \leq \Pr\{\text{Bin}(K, p_k) \geq k\} = \Pr\{\text{Beta}(k, K + 1 - k) \leq p_k\} = \delta$. $\square$

The case $p > q$

The constant of § 4.1 is

$$C_{p,q} = \sup_{\alpha_L + \alpha_R = 1 - p} \inf_{\beta_L + \beta_R = 1 - q} \max(c_{1 - \alpha_L, 1 - \beta_L}, c_{1 - \alpha_R, 1 - \beta_R}).$$

Numerically the worst split puts the whole budget on one end, so that $C_{p,q} = c_{p,q}$; this observation is not used. Ball arithmetic gives $-0.0572 \leq C_{0.9036, 0.9} \leq -0.057$, where 0.9036 is the order-statistic level for $K = 110$, $k = 105$ at $\delta = 0.04$, used in § S4.

In the proof of the upper bound of Theorem 1, replace $1 - q$ by $1 - p$ for the noisy budget, choose any split $\beta_L + \beta_R = 1 - q$ of the latent budget, and use $F_Y^{-1}(1 - \beta_R) \leq c_{1 - \alpha_R, 1 - \beta_R}$ at each end. Since $c_{p',q'}$ decreases in p' , the worst case has $\alpha_L + \alpha_R = 1 - p$, which gives $R_{p,q}(x) \leq 1 + C_{p,q}x^{1/2}$ for every x , with $C_{p,q}$ nonincreasing in p . The proof of the lower bound applies with $b_u(q)$ replaced by $b_u(p)$ and $p' = p + \eta_x$, giving $R_{p,q}(x) \geq 1 + c_{p,q}x^{1/2} - o(x^{1/2})$ for every $p \geq q$. This proves Corollary 1, whose numerical intervals are certified in § S3.

Bi-log-concave laws

Corollary S1. *Let $\mathcal{B}$ be the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave (Dümbgen et al., 2017); $\mathcal{L} \subset \mathcal{B}$. Define $R_{p,q}^{\mathcal{B}}$ by (1) with $\mathcal{B}$ in place of*

$\mathcal{L}$. Then Lemma 1 holds over $\mathcal{B}$ with the same $c_{p,q}$, and for $p \geq q$ and every $x > 0$, $R_{p,q}^{\mathcal{B}}(x) \leq 1 + C_{p,q}x^{1/2}$, while $R_{q,q}^{\mathcal{B}}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$. In particular Corollary 1 holds over $\mathcal{B}$.

Proof. The proof of Lemma 1 uses only the concavity of $\log F_Y$ at the q -quantile, so its upper bound holds for every Y with $\log F_Y$ concave, and exponential tails are in $\mathcal{L}$, which gives equality. In the proof of Theorem 1, $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave, and $(-W - 1)/x^{1/2}$ has a concave log distribution function because $\log(1 - F_W)$ is concave and F_W is continuous. The lower bound uses laws in $\mathcal{L} \subset \mathcal{B}$. $\square$

The class $\mathcal{B}$ contains multimodal laws. Proposition 1 and the box-bound upper bounds of § S3 use log-concavity of the density; the ball-arithmetic constants do not (Corollary S1). Theorems 2 and 3, Lemma 2, Proposition 4 and Corollary 2 hold over $\mathcal{B}$: their proofs use the monotone hazards of $\mathcal{B}$, the bound $f' \leq f^2/F$ of Dümbgen et al. (2017), and the preservation of $\mathcal{B}$ under convolution with a log-concave law and under weak limits (Saumard, 2019).

Symmetry alone does not make noisy calibration conservative in $\mathcal{B}$. The density $f(w) = \{1 + 0.05 \cos(2\pi w)\}/2$ on $(-1, 1)$ is in $\mathcal{B}$, because $\sup |f'| = 0.05\pi < \inf f^2 = 0.2256$ (Dümbgen et al., 2017, Theorem 1), but it is not unimodal. At $q = 0.7$, $r_0 = Q_q(|W|) \approx 0.7077$ and $f'(r_0) > 0$; since f' is odd, $\partial_s \text{pr}(|W + s^{1/2}Z| \leq r_0) = f'(r_0) > 0$ at $s = 0$, so noise raises the coverage and the noisy quantile falls below r_0 , to 0.707676 at $s = 10^{-5}$, against $r_0 = 0.707678$. Anderson’s comparison needs unimodality. The bimodal law of Table 1, the mixture $0.4N(m_1, s^2) + 0.6N(m_2, s^2)$ with mean zero, variance one and $(m_2 - m_1)/s = 2.5$, is in $\mathcal{B}$ by the same criterion, $f'F \leq f^2$ and $-f'(1 - F) \leq f^2$, checked on a grid of spacing 4×10^{-5} on $[-8, 8]$ (test suite); with $p = 0.4$ the mixture stays in $\mathcal{B}$ up to a separation of about 2.59, where $\max(f'F - f^2)/f^2$ changes sign, while bimodality starts at about 2.443; at 2.5 the largest value of $(f'F - f^2)/f^2$ and of $\{-f'(1 - F) - f^2\}/f^2$ is -0.036 . The parameters are $m_1 = -0.94868$, $m_2 = 0.63246$ and $s = 0.63246$. The bound of Theorem 2 with $E(W) = 0$ holds for this law and its version with the cosine phase-shifted by $\pi/2$ and recentred to mean zero, at $q = 0.3, 0.7, 0.9$ and $s = 10^{-3}, 0.1, 1$ (test suite).

S2 The shape of the sharp radius

Figure S1 shows $R_{q,q}(x) - 1$ computed by grid maximization over the extremal family; grid values bound $R_{q,q}$ from below. The levels at which the grid value crosses 1 are $x^*(q) \approx 0.034, 0.017, 0.0062$ and 0.0012 for $q = 0.8, 0.85, 0.9$ and 0.95 , and the largest widening found is 0.37%, 0.17%, 0.063% and 0.012%. At $q = 0.9$ the numerical upper bounds of § S3 give $R_{q,q}(x) \leq 1$ for $x \geq 0.013$, so Theorem 1 bounds the widening by $c_{0.9} 0.013^{1/2} < 0.22\%$ at every x ; this rests on the double-precision bounds, whereas the bound 1.16% of § 3 rests only on Theorem 1 and the certified $c_{0.9}$. Below $x^*(q)$ the noisy threshold must be widened; above it, grid values below 1 do not by themselves show that shrinking is safe, which the numerical upper bounds of § S3 do. The exact grid values lie below $1 + c_{0.9}x^{1/2}$ at $x = 10^{-6}, \dots, 10^{-2}$, as Theorem 1 requires.

Centred laws and the transition

The following values are computed in double precision by `experiments/e32_centering.py` and are not certified. For $q = 0.8, 0.9$ and 0.95 : $\kappa_q^* = 6.8118, 20.1823$ and 50.0741 , attained at $u^* = 0.11328, 0.044286$ and 0.018943 . The two upper bounds in Fig. 1 cross at $\kappa = 10.829, 26.221$ and 59.466 , where L_q/U_q is smallest, 0.770, 0.879 and 0.927; as $\kappa \rightarrow \infty$ the ratio tends to $1 - \log(1/q) = 0.777, 0.895$ and 0.949 . At $q = 0.9$, $L_q(25) = 0.017393$ and $L_q(50) = 0.0089448$. For the mean-zero laws of Proposition 2, the coefficient with $\beta = 0.2$ and $\ell = 2$ at $q = 0.9$ is 0.0156540; the expansion is accurate to 6×10^{-5} at $x = 1.3 \times 10^{-4}$ but fails at $x \approx 1.1 \times 10^{-3}$, where the noise reaches the end of the support and the threshold may already be shrunk. The bound of Theorem 2, including its form with $|E(W)| \leq B$, was checked on 642 laws from the family of Proposition 2, with no violation.

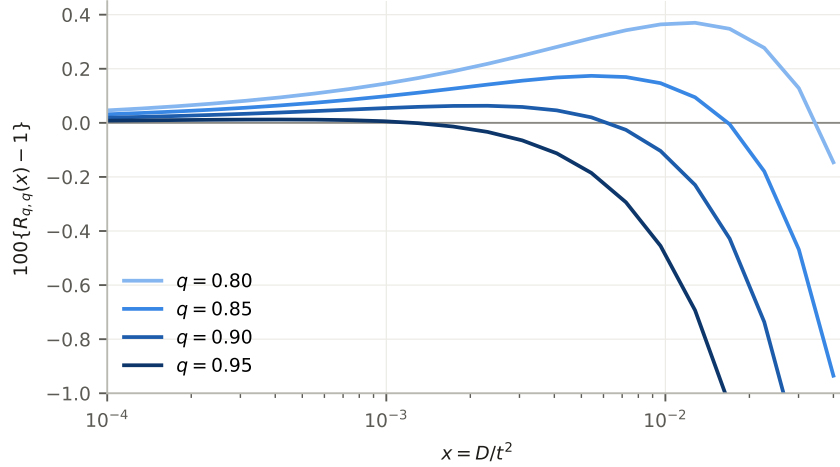


Figure S1: Sharp latent radius relative to the noisy threshold, $100\{R_{q,q}(x) - 1\}$, for four coverage levels, computed by grid maximization over the extremal family, which bounds $R_{q,q}$ from below.

The location of the mean in examples

Figure S2 shows the radius correction for laws scaled so that the noisy coverage of $[-1, 1]$ is exactly 0.9, computed from closed forms by `experiments/e35_mean_location.py`. For the boundary-layer law $W = 1 + x^{1/2}\{b - E_u\}$ at the maximizer u^* of v_q , whose mean lies $20.18x^{1/2}$ from the boundary, the correction divided by $x^{1/2}$ is 0.019062 for $x \leq 3 \times 10^{-5}$, equal to $c_{0.9}$. For the same family at the rate with $k_q(u) = 50$ it is $0.008945 = L_{0.9}(50)$. For the fixed mean-zero law of Proposition 2 with $\beta\ell = 0.5$ on a unit segment, the correction divided by x is 0.024005 up to $x = 5 \times 10^{-5}$, and it turns negative once the noise reaches the end of the support. Larger centred coefficients sit closer to that end: at $\beta\ell = 0.9$ the coefficient is 0.070 but the quantile lies 0.0011 from the end, and the correction is positive only for x below about 10^{-7} . A Gaussian law with mean $1.29x^{1/2}$ from the boundary after scaling, $W \sim N(1 - 2x^{1/2}, 0.01x)$ before scaling, needs no correction at all: its radius is $1.16x^{1/2}$ below the noisy threshold. A mean near the boundary is necessary for a correction of order $x^{1/2}$, not sufficient.

The limit with κ fixed

The limit in Theorem 3 is taken with κ fixed. Applying the small- u construction to a centred law, that is, $\kappa = x^{-1/2}$, gives in the limit a left exponential tail of rate $1 - \log(1/q)$ whose noise-free central coverage is $q\{1 - e^{-2+2\log(1/q)}\}$, 0.750 at $q = 0.9$, far below the constraint. So $\{1 - \log(1/q)\}/2$ is not a lower bound for the centred coefficient.

S3 Certification

Box bounds for the radius

By the symmetry $W \mapsto -W$ of the kernel and of $|W|$ it suffices to treat $\beta \geq 0$. Write a member of $\mathcal{F}$ with $\beta \geq 0$ as $W = b - Y$, where Y has density proportional to $e^{-\beta y}$ on $[0, \ell]$. The ends of the family are included: $\ell = 0$ or $\beta = \infty$ gives a point mass and $\ell = \infty$ an exponential tail. For slopes $\beta' > \beta$ the likelihood ratio $e^{-(\beta' - \beta)y}$ is decreasing, so $Y_{\beta'}$ lies below Y_{β} in likelihood-ratio order, and for lengths $\ell < \ell'$ truncation gives $F_{\ell} \geq F_{\ell'}$. Thus W is stochastically increasing in β and b and decreasing in ℓ . On a box of parameters let

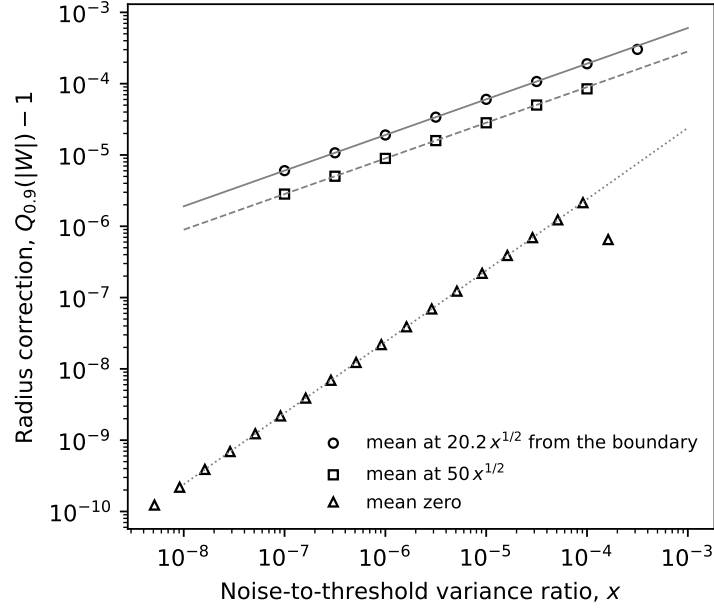


Figure S2: Radius correction $Q_{0.9}(|W|) - 1$ against x on logarithmic axes, for laws with noisy coverage exactly 0.9: mean at $20.2x^{1/2}$ from the boundary (circles), at $50x^{1/2}$ (squares) and mean zero (triangles), with $c_{0.9}x^{1/2}$ (solid), $L_{0.9}(50)x^{1/2}$ (dashes) and $0.024x$ (dots). Only positive corrections are shown.

W_{lo} and W_{hi} be the stochastically smallest and largest corners. Since $\text{pr}(W + e \leq \pm 1)$ and $\text{pr}(W \leq y)$ are expectations of decreasing functions of W , every law in the box satisfies

$$\text{pr}(|W + e| \leq 1) \leq \text{pr}(W_{\text{lo}} + e \leq 1) - \text{pr}(W_{\text{hi}} + e \leq -1), \quad \text{pr}(|W| \leq s) \geq \text{pr}(W_{\text{hi}} \leq s) - \text{pr}(W_{\text{lo}} \leq -s).$$

The density bound $\text{pr}(|W + e| \leq 1) \leq 2 \sup f_W = 2\beta/(1 - e^{-\beta\ell})$ is also available. A box whose first bound is below p , or whose second bound is at least q , contains no law that violates $R_{p,q}(x) \leq s$, and a branch and bound over boxes certifies $R_{p,q}(x) \leq s$. The location is confined to $[-1 + x^{1/2}\Phi^{-1}(p), b_{\max}]$: the lower end holds because $W \leq b$; for the upper end take $k = 7x^{1/2}$; the density of W increases towards b , so $\text{pr}(|W| \leq 1 + k) \leq r/(1 + r)$ with $r = (2 + 2k)/(b - 1 - k)$, and $\text{pr}(|e| > k) = 2\Phi(-7)$. The parameters $\beta, \ell \in [0, \infty]$ are compactified.

Proposition S1. For all $x, h \geq 0$, $R_{p,q}(x + h) \leq R_{p,q}(x) + c_q h^{1/2}$.

Proof. If W is feasible at noise level $x + h$, then $W + h^{1/2}Z'$, with $Z' \sim N(0, 1)$ independent of (W, Z) , is log-concave and feasible at level x . So $s = Q_q(|W + h^{1/2}Z'|) \leq R_{p,q}(x)$, and Theorem 1 at scale s gives $Q_q(|W|) \leq s + c_q h^{1/2}$. $\square$

We computed numerical upper bounds of $R_{p,0.9}$ for $p = 0.9, 0.9036$ and 0.9068 on $0.002 \leq x \leq 0.366$ with step 5×10^{-4} , using Proposition S1 between grid points. Here 0.9068 and 0.9036 are the order-statistic levels p_k of $K = 110$, $k = 105$ at $\delta = 0.05$ and $\delta = 0.04$; the latter is used in § S4. The closed forms agree with 30-digit quadrature to within 10^{-11} at random points with $10^{-4} \leq x \leq 0.37$, $10^{-4} \leq \ell \leq 20$ or $\ell = \infty$ and $10^{-3} \leq \beta\ell \leq 10^3$, which cover the ranges in which boxes are evaluated after outward rounding (test suite; the largest error found was 1.1×10^{-12}), and boxes are cleared with a margin of 10^{-9} ; the bounds therefore rest on double-precision evaluation of the closed forms and are not interval arithmetic. For $0.05 \leq x \leq 0.3$ the bound exceeds a feasible value by at most a relative 3×10^{-4} for $p = 0.9$ and 10^{-3} for $p = 0.9068$. The

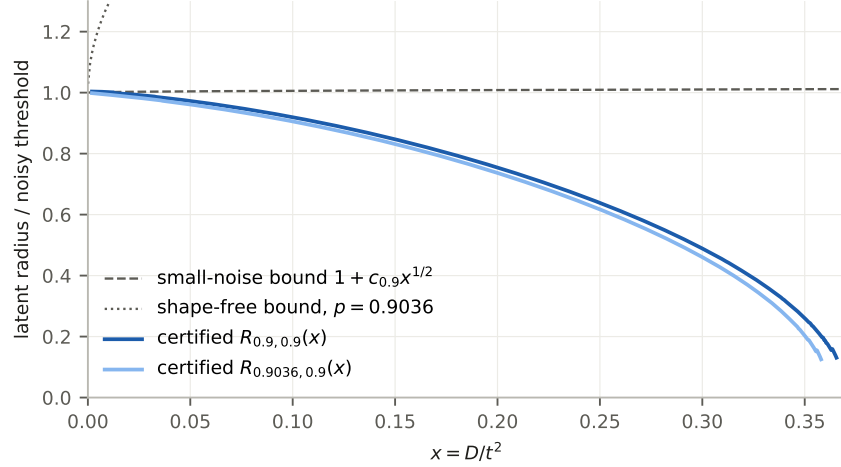


Figure S3: Numerical upper bound of the sharp radius $R_{p,0.9}(x)$ for $p = 0.9$ and for the order-statistic level $p = 0.9036$ ($\delta = 0.04$), with the small-noise bound of Theorem 1 and the shape-free bound $1 + x^{1/2}z_{1-(p-q)/2}$, which follows from $|W| \leq |V| + |e|$.

branch and bound fails only just before the feasibility edges, $x \approx 0.370$ for $p = 0.9$ and 0.361 for $p = 0.9036$; there Proposition S1 from the last grid point gives $R_{0.9,0.9}(x) \leq 0.132$. Below $x = 0.01$ the lookup also uses Theorem 1 and its $p \geq q$ form, which are within 0.3% of a feasible value there (Fig. S3). At $q = 0.9$ the numerical bounds fall below 1 from $x = 0.013$ on, with the continuity margin $c_{0.9}(5 \times 10^{-4})^{1/2} \approx 4.3 \times 10^{-4}$ keeping them below 1 between grid points, and reach 0.973 at $x = 0.05$. A differential-evolution search over the three-parameter family, which evaluates the constraint by Gauss–Legendre quadrature, returned 1.027 at $x = 0.001$, above the bound 1.0006 of Theorem 1: the quadrature is inaccurate for steep slopes at small x . That search is therefore used only for $x \geq 0.006$, and the legacy table `results/r_table.json` built with it is not used in the paper.

Ball-arithmetic certificates for the constants

Every deciding inequality is evaluated in Arb ball arithmetic; floating point only proposes points, balls are converted to doubles with outward rounding, and sums of bounds are formed as balls.

1. $m_u(b)$ decreases in b , and in u because E_u is stochastically decreasing in its rate. Hence $b_u(p)$ is nonincreasing in u , and on $[u_0, u_1]$, $b_u(p) - u^{-1} \log(1/q) \leq b_{u_0}(p) - u_1^{-1} \log(1/q)$. A ball with $m_u(b) < p$ proves $b_u(p) < b$.
2. For $u \geq u_{\max}$ the value is at most $b_{u_{\max}}(p)$. For small u , let $b(u) = \{\log(1/p) + u^2/2 - \log(1 - u^3)\}/u$. Then $e^{-ub+u^2/2} = p(1 - u^3)$, and the Mills ratio gives $\Phi(-b) \leq pu^3$ whenever $\phi\{\log(1/p)/u\} \leq pu^2 \log(1/p)$. This holds on $(0, u_{\min}]$ once it holds at $u_{\min} \leq \log(1/p)/2^{1/2}$, because the left side divided by u^2 increases there. So $b_u(p) \leq b(u)$, and for $p \geq q$ the resulting bound increases in u .
3. By Lemma 1, $c_{p',q'} \leq \gamma$ if and only if $1 - q' \geq \varphi_\gamma(1 - p')$, where $\varphi_\gamma(a) = \max[0, \sup_u 1 - \exp\{-u(b_u(1 - a) - \gamma)\}]$. Hence $C_{p,q} \leq \gamma$ if $\varphi_\gamma(a_L) + \varphi_\gamma(a_R) \leq 1 - q$ whenever $a_L + a_R = 1 - p$. Since φ_γ is nondecreasing, it suffices to check this at the ends of an adaptive partition of a_L , with φ_γ bounded above by the same monotone branch and bound in u .

The certificates give $c_{0.8} \in [0.0460730, 0.0460731]$, $c_{0.9} \in [0.0190618, 0.0190619]$, $c_{0.95} \in [0.0084075, 0.0084077]$ and $c_{0.99} \in [0.0013950, 0.0013958]$, $-0.0571845 \leq c_{0.9036,0.9} \leq C_{0.9036,0.9} \leq -0.057$, $-0.1149813 \leq$

$c_{0.9068,0.9} \leq C_{0.9068,0.9} \leq -0.114$, and $C_{q+\delta,q} \leq 0$ for $(q, \delta) = (0.8, 0.005)$, $(0.9, 0.001)$ and $(0.95, 0.0002)$. Lower bounds $c_{q+\delta,q} > 0$ at $\delta = 0.0043$, 0.00079 and 0.00015 come from single points u , evaluated in ball arithmetic, and $C_{p,q} \geq c_{p,q}$. Double-precision root finding puts the smallest slack at 4.4×10^{-3} , 8.0×10^{-4} and 1.6×10^{-4} .

S4 Heterogeneous and estimated noise variances

Heterogeneous known variances

In the heterogeneous rule, ε bounds, for each data set, the sup-norm error of the average kernel after compressing the D_i to 32 support points: the maximum over a grid of spacing $h = 10^{-3}$ plus $Mh^2/8$, with M a bound on the second derivative; it was about 2×10^{-4} . For each data set a numerical upper bound of R^{mix} is computed by the box bounds, which hold term by term for a mixture kernel, and the shape-free bound is used if the branch and bound fails.

Replacing the average kernel by the kernel of a single variance is not safe. With the mean variance plugged in, the radius is 1.4% too short at mean $x = 0.2$ with lognormal spread 1 (numerical). The smallest variance also fails: take $W = b - E_2$ with $b = 1.0435647$ and noise an equal mixture of $N(0, 10^{-8})$ and $N(0, 5 \times 10^{-4})$. The noisy coverage of $[-1, 1]$ exceeds 0.9, while the radius $1 + c_{0.9}10^{-4}$ given by the smaller variance covers W with probability 0.89977; this was checked in 192-bit ball arithmetic.

Estimated variances

Suppose the D_i are unknown and estimates $\hat{D}_i$ are available that are independent of the scores, as for the sample means and variances of normal unit data. The order-statistic event of Proposition 3 involves the scores only. Let $\mathcal{E}_\eta$ be an event determined by the $\hat{D}_i$ with $\text{pr}(\mathcal{E}_\eta) \geq 1 - \eta$.

Lemma S1. *Let $G \in \mathcal{L}$. Suppose that on $\mathcal{E}_\eta$ a continuous kernel u_t , vanishing at $\pm\infty$, satisfies $u_t \geq \bar{g}_{D,t}$ pointwise for every $t > 0$. Then $s = T R_{p_k,q}[u_T]$, the radius (1) with kernel u_T , covers a new latent residual with probability at least q , conditionally on the data, with probability at least $1 - \delta - \eta$. If $\mathcal{D}_\eta$ is a confidence set with $\text{pr}(D \in \mathcal{D}_\eta) \geq 1 - \eta$, the same holds for $s = T \sup_{d \in \mathcal{D}_\eta} R_{p_k,q}^{\text{mix}}(d/T^2)$. With a compressed kernel, p_k is replaced by $p_k - \varepsilon$ as in § 4.2.*

Proof. On both events, $\int u_T dG \geq \int \bar{g}_{D,T} dG \geq p_k$, and Proposition 1 applies to u_T . The union bound gives $1 - \delta - \eta$, with no independence between the two events needed. $\square$

Lemma S2. *For $\lambda \geq 1$, $R_{p,q}^{\text{mix}}(\lambda x_1, \dots, \lambda x_K) \leq \lambda^{1/2} R_{p,q}^{\text{mix}}(x_1, \dots, x_K)$.*

Proof. If W is feasible at λx , then $U = \lambda^{-1/2}W$ satisfies the constraint at threshold $\lambda^{-1/2} \leq 1$, hence at threshold 1, so $Q_q(|W|) = \lambda^{1/2}Q_q(|U|) \leq \lambda^{1/2}R_{p,q}^{\text{mix}}(x)$. $\square$

A confidence box for all K variances needs level $1 - \eta/K$ per area and is useless with one or two degrees of freedom. Survey practice instead models variances through a few parameters, as generalized variance functions do. In the simplest such model $D_i = \sigma^2 a_i$ with known a_i , for instance $a_i = 1/n_i$, and the pooled estimate satisfies $\nu \hat{\sigma}^2 / \sigma^2 \sim \chi_\nu^2$ with $\nu = \sum_i (n_i - 1)$. The confidence set is the segment $\{\sigma^2 a : \ell \leq \sigma^2 \leq h\}$. Lemma S2 covers the supremum over it by finitely many points without assuming that R is monotone in σ^2 : for $\ell = s_0 < \dots < s_m = h$,

$$\sup_{\ell \leq \sigma^2 \leq h} R_{p_k,q}^{\text{mix}}(\sigma^2 a / T^2) \leq \max_j (s_{j+1}/s_j)^{1/2} U_j, \quad U_j \geq R_{p_k,q}^{\text{mix}}(s_j a / T^2).$$

Plugging in the lower end ℓ alone would not be justified, since R can increase with the noise level (Theorem 1).

In simulations, areas have n_i normal units, so $D_i = \sigma_i^2/n_i$ and the within-area variances are independent of the direct estimates; $K = 110$, mean $D = 0.577$, and 100 replications per setting and law (normal, Laplace, truncated Laplace). The rule uses $\delta = 0.04$ and $\eta = 0.01$, split as 0.009 below and 0.001 above the pooled interval for σ^2 , and needs 18 grid points on average. The radii below use converged grid values at those points; for the first 30 data sets of every setting and law (270 in all) the rule was rerun with numerical upper bounds U_j at every grid point, as Lemma S1 requires. The branch and bound succeeded for every data set, and the half-width grew by 0.14% on average and by at most 1.0%, at about 3.6 minutes per data set on one core.

1. With $n_i \in \{2, 3\}$, $\nu \approx 165$, the rule is 6.3–6.5% shorter than noisy conformal calibration and 2.9–3.0% longer than the known-variance rule at the same level, which is 9.0–9.1% shorter than noisy conformal calibration. With $n_i \in \{5, \dots, 10\}$, $\nu \approx 740$, the figures are 7.1–7.7% and 1.6–1.7%. Relative to the known-variance rule at $\delta = 0.05$, which needs no budget for the variances and meets the same final target, 62–63% and 73–74% of its gain survive. Reliability of the rule with grid values was at least 0.98; with numerical upper bounds, on the 30 data sets per setting, it was at least 0.967.
2. The supremum over the interval exceeded the value at its lower end by 0.2%.
3. When the within-area variances differ, with lognormal spread 0.5, the common-variance model is wrong and no guarantee applies; the rule was still reliable, at least 0.99, and 5.9–6.5% shorter than noisy conformal calibration.
4. Without pooling, with each area’s variance estimated from ν own degrees of freedom, a kernel envelope built from area-wise intervals and a binomial bound on the number of misses is 31–40% longer than noisy conformal calibration at $\nu = 2$, within 1.3% of it at $\nu = 50$, and 4.7–5.0% and 6.6–7.2% shorter at $\nu = 200$ and 1000 (30 replications per law; grid values). Whether a better area-wise construction exists is open.

S5 The shape-free comparison rule

Proposition S2. *Let $\bar{g}$ be a kernel that is symmetric, continuous, strictly decreasing in $|w|$ and tends to 0 as $|w| \rightarrow \infty$, with $m = \bar{g}(0)$, and suppose that $qm < p < m$. Over all laws G on the real line with $\int \bar{g} dG \geq p$, the supremum of $Q_q(|W|)$ is the radius r that solves $\bar{g}(r) = (p - qm)/(1 - q)$.*

Proof. Let $\tau = (p - qm)/(1 - q) \in (0, m)$. Since $m - \bar{g}(W) \geq 0$, Markov’s inequality gives $\text{pr}\{\bar{g}(W) < \tau\} \leq \{m - p\}/(m - \tau) = 1 - q$, and $\bar{g}(W) < \tau$ exactly when $|W| > r$, so $Q_q(|W|) \leq r$. Conversely, for small $\epsilon > 0$ put mass $q - \epsilon$ at 0 and mass $1 - q + \epsilon$ at the point $w_\epsilon > 0$ with $\bar{g}(w_\epsilon) = \{p - (q - \epsilon)m\}/(1 - q + \epsilon)$. This law meets the constraint with equality, has $Q_q(|W|) = w_\epsilon$, and $w_\epsilon \rightarrow r$ as $\epsilon \rightarrow 0$. $\square$

The average kernel of § 4.2 satisfies these conditions, so for a fixed rank the shape-free rule of Table 1 is the smallest radius that the single order-statistic constraint allows without a shape assumption. With 1 in place of $\bar{g}(0)$ it contains the bound of Guille-Escuret & Ndiaye (2024). It is not a bound on procedures that use several thresholds or tune the rank on independent data, as LatentCP (Zheng et al., 2026) can.

S6 Simulation details and an application

Table 1

Conditional coverage is computed from closed-form latent distribution functions. With 150 data sets the binomial standard error of a reliability is 0.018 at 0.95 and at most 0.029 at the lowest reliability observed,

0.847; for instance $145/150 = 0.967$ has the 95% binomial interval $[0.924, 0.989]$. The computed radius uses the order-statistic level $p_k = 0.9068$ of $\delta = 0.05$. The branch and bound returned a numerical upper bound of R^{mix} for every one of the 1050 data sets, at relative tolerance 10^{-3} in 671, 3×10^{-3} in 363 and 10^{-2} in 16, so the shape-free fallback was never used. With the mean variance in a single Gaussian kernel, which is not a valid plug-in, the numerical radius at level p_k is 2.9–3.0% shorter than at level $p = q$ (LDC_cert_pk against LDC_PAC in `results/hetldc_synth_summary.csv`), so most of the gain over a $p = q$ calibration comes from the order-statistic level. The shape-free rule was run at every fixed rank 106–110 and all results are stored; rank 107 is reported because it is best on average over the first four laws, which favours the comparison rule. The Fay–Herriot interval falls short of the target under the Laplace law by about 1.3 standard errors and under the truncated Laplace law, with reliability 0.847, by about 5.8. The Fay–Herriot tolerance interval is our adaptation of the parametric bootstrap of Chatterjee et al. (2008) to an intercept-only model: the bootstrap $(1 - \delta)$ -quantile of the multiplier each refit needs to cover $N(\hat{\mu}, \hat{A})$ with probability 0.9. The latent residuals are drawn directly, with no fitted centre, and the mean noise variance 0.577 matches the ratio of design to model variance in the application below.

Reliability of the analytic rules with 2000 data sets

The noisy threshold, Corollary 2 and the Markov rule need no numerical optimization, so `experiments/e36_analytic_reli` reran them on 2000 fresh data sets for each of the seven laws of Table 1. Every rule reached conditional coverage 0.9 in all 2000 data sets for every law, so each reliability has Clopper–Pearson 95% interval $[0.998, 1]$, including the t_3 law, outside $\mathcal{B}$. Mean conditional coverage was 0.969–0.992 for the noisy threshold, 0.968–0.991 for Corollary 2 and 0.984–0.999 for the Markov rule, and the 5% quantile of conditional coverage was at least 0.947, 0.945 and 0.968. The guarantees are therefore conservative in these designs: the latent laws are far from the extremal laws of § 3.

Oracle-matched widths

In a larger experiment with seven latent shapes, common and heterogeneous noise and 2000 data sets per setting, each rule’s width was rescaled by a single factor, chosen with knowledge of the latent law, to reach reliability 0.95 exactly. At matched reliability, marginal noisy conformal calibration was shortest in 12 of the 14 settings and the Fay–Herriot tolerance interval in the other 2; relative to noisy conformal calibration the Fay–Herriot tolerance interval was 3.1% shorter to 12.1% longer and the log-concave rule 4.5–11.5% longer (`results/conditional_synth_exact_matched.csv`). The log-concave rule with its guarantee was a median 15% (range 9–33%) wider than its own oracle-matched width, and the Fay–Herriot tolerance interval a median 6% (range –3 to 16%). So the advantage of the log-concave rule over noisy conformal calibration lies in obtaining a latent guarantee cheaply, not in a better interval shape.

Comparison with LatentCP and deconvolution

`experiments/e34_competitors.py` runs the following rules on the 600 data sets of the first four laws of Table 1; widths are relative to the computed radius. For the other three laws only the two analytic rules were run.

1. LatentCP (Zheng et al., 2026), single level with $\gamma = \alpha/2$, has a marginal guarantee. Its latent set depends on the new unit’s noise variance, which an unsampled area does not have; we draw it from the distribution that generated the D_i , 400 draws per data set, and average coverage and width over the draws. It was 16–18% longer, with mean coverage 0.976–0.990 and reliability 1.0. The tuned single-level and two-level versions, tuned on half the data, were 15–25% longer. A PAC version with a binomial rank, our adaptation, was 39–58% longer.

2. The deconvolution threshold of Cohen et al. (2025), with the mixture kernel in place of their homoscedastic kernel, collapses at $K = 110$ with their bin width 0.01: most bins are empty and the masked fit gives mean coverage 0.32–0.44. With bin width 0.2 and the empty bins fitted as zeros it had mean coverage 0.88–0.90, reliability 0.45–0.67 and width 22–30% shorter; it has no latent guarantee.
3. The noisy threshold of Proposition 4 was 10.5–11.7% longer and Corollary 2 9.1–10.2% longer, both with reliability 1.0.

The tuning grids of LatentCP and the scan step 0.01 of the deconvolution threshold are our choices; the original papers do not fix them for this setting.

School districts

In the finite population `apipop` (Lumley, 2004), 110 of 325 school districts are used for training and 110 for calibration, with 2 or 3 schools sampled per district. Coverage is the share of the remaining 105 districts that are covered. Design variances are estimated with 1–2 degrees of freedom and plugged in as known by every rule, a case that no guarantee covers. Over four conditions of 120 replications, the reliability of the heterogeneous rule was 0.942–0.992, of the mean-variance rule 0.983–1.000 and of the Fay–Herriot interval 0.908–0.992, below target in two conditions. The heterogeneous rule at level p_k was 0.8–5.2% shorter than the mean-variance rule with the $p = q$ table, mostly because of the level, and 10–16% longer than the Fay–Herriot interval.

S7 Reproducibility

All computations are in the public repository <https://github.com/kunhailP/noisy-calibration-latent-coverage>. The core routines are in `src/uai`: closed forms and the transition coefficient in `extremal.py`, the box certificates in `certify.py`, ball arithmetic in `interval.py`, and the procedures in `procedures.py` and `estimated.py`. The test suite, run by `make test`, checks the constants, the counterexamples and the centring bounds, and the `Makefile` lists the command that produces each group of results. The experiments are numbered scripts in `experiments`; `e27` produces the ball-arithmetic certificates, `e24` and `e25` the certified tables, `e20` Table 1, `e29` and `e30` the estimated-variance results, `e31` the shape-free rule, `e32` the centring and transition values, `e34` the comparison with LatentCP and deconvolution, `e35` and `fig_mean_location.py` Fig. S2, and `fig_transition.py` Fig. 1. The analytic rules of Proposition 4 and Corollary 2 are `noisy_threshold_halfwidth` and `simple_shrink_halfwidth` in `procedures.py`; `hetldc_certified` gives the computed radius, while `hetldc_halfwidth` returns a grid value, a lower bound of R^{mix} , and is not a guaranteed radius.

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